

Oleg S. Pinykh

Digital Imaging and Communications in Medicine (DICOM)

Oleg S. Pianykh

Digital Imaging and Communications in Medicine (DICOM)

A Practical Introduction
and Survival Guide

With 96 Figures and 57 Tables

Oleg S. Pinykh, Ph.D.

Department of Radiology, BIDMC

Harvard Medical School

330 Brookline Ave

Boston, MA 02215

USA

ISBN 978-3-540-74570-9

e-ISBN 978-3-540-74571-6

DOI 10.1007/978-3-540-74571-6

Library of Congress Control Number: 2007943072

© 2008 Springer-Verlag Berlin Heidelberg

This work is subject to copyright. All rights are reserved, whether the whole or part of the material is concerned, specifically the rights of translation, reprinting, reuse of illustrations, recitation, broadcasting, reproduction on microfilm or in any other way, and storage in data banks. Duplication of this publication or parts thereof is permitted only under the provisions of the German Copyright Law of September 9, 1965, in its current version, and permission for use must always be obtained from Springer Verlag. Violations are liable to prosecution under the German Copyright Law.

The use of general descriptive names, registered names, trademarks, etc. in this publication does not imply, even in the absence of a specific statement, that such names are exempt from the relevant protective laws and regulations and therefore free for general use.

Product liability: The publishers cannot guarantee the accuracy of any information about dosage and application contained in this book. In every individual case the user must check such information by consulting the relevant literature.

Cover design: Frido Steinen-Broo, eStudio, Calamar, Spain

Typesetting and Production: le-tex publishing services oHG, Leipzig, Germany

Printed on acid-free paper

9 8 7 6 5 4 3 2 1

springer.com

To my parents

Foreword

Digital Imaging and Communications in Medicine in Radiology

The DICOM (Digital Imaging and COmmunications in Medicine) standard is the backbone of modern image display, equivalent only to film in the predigital era. Since the inception of this standard some 20 years ago, it has become the driving force behind the entire imaging workflow. DICOM truly controls all parts of digital image acquisition, transfer and interpretation, and many radiologists and other imaging specialists and users may not realize to what extent their work relies on DICOM capabilities.

We depend on DICOM extensively. We use DICOM to collect the original medical images. This process is much more complex compared to X-Ray film, but is also more accurate and complete. We use DICOM to send, distribute, and store images, irrespective of machine, manufacturer, or modality. DICOM controls proper image display, and without DICOM we wouldn't have any image postprocessing – be it simple multiplanar reconstruction, or more advanced perfusion analysis, virtual colonoscopy, volume segmentation, or computer-aided diagnosis. We owe to DICOM much of the ease and flexibility that we enjoy in our work.

As digital medicine becomes increasingly complex and imaging projects globalize, knowledge of DICOM basics for any healthcare professional becomes crucial. Whether you are thinking about your next picture archiving and communication system upgrade, or a new teleradiology project, or simply about installing a new digital modality, DICOM should be the crucial reference. Standard, DICOM-based workflow is the only way to build a robust and efficient radiology practice. It is also the only way to integrate your medical imaging into a complete enterprise-wide electronic patient record solution.

This book, written from a hands-on angle, gives the healthcare professional a comprehensive introduction to the evolving and multifaceted standard that is DICOM. It may become your daily reference, a teaching guide, or a tool to prepare you for dealing with the intricacies of original DICOM volumes and supplements. Take full advantage of it, and take full advantage of DICOM.

Vassilios Raptopoulos, M.D.
Interim Radiologist-in-Chief,
Beth Israel Deaconess Medical Center,
Boston, USA
Professor of Radiology,
Harvard Medical School, USA

Foreword

Digital Imaging and Communications in Medicine in Healthcare Information Technology

Unlike healthcare standards such as HL7, the DICOM (Digital Imaging and COMmunications in Medicine) standard is not as well known or understood by information technology (IT) professionals. On one hand, DICOM stands and delivers on the promise of interoperability and cost effectiveness. On the other hand, it requires some tweaking to work. DICOM planning and implementation errors are common, often resulting in operational nightmares. Given that digital medicine and picture archiving and communication systems (PACS) require a substantial investment and are critical, getting the DICOM part right up front is essential.

Frequently overlooked is the impact of DICOM and PACS on the actual patient. DICOM drives clinical workflow from the moment a patient enters the radiology department, collecting important imaging data from the digital modality and delivering it to the radiologist in the most accurate and diagnostically complete form within minutes. Comparing previous studies with the current study is effortless with DICOM, resulting in more skillful diagnosis and treatment. No film to lose, misplace, or under- or overexpose, and fewer repeated studies all add up to faster and vastly improved healthcare. DICOM is beneficial to the patient, provider, and the facility and IT professionals owe it to them to get it right. The importance of DICOM as a working, integral part of the IT infrastructure will continue to expand. The number of present and available DICOM and PACS applications is staggering. Selecting the PACS right for you becomes more challenging with each passing year and a good understanding of DICOM is now essential.

Without a basic and practical understanding of DICOM you risk making embarrassing mistakes in such areas as security, equipment specification, capacity planning, network configuration, and disaster recovery/continuity of operations.

The built in security of DICOM is not going to be adequate to address Health Insurance Portability and Accountability Act (HIPAA) requirements and will need to be augmented. The various modalities have different image resolutions, which affect the choice of viewing monitors. DICOM supports several image-compression technologies. The tradeoff between image sizing and quality will affect equipment sizing and network bandwidth requirements. The DICOM connectivity model is still very much static. Remote access through a virtual private network (VPN) needs to be carefully planned. The complexity and interactions of all of the above are interrelated and can make upgrades difficult. The aspects of disaster recovery and continuity of operations emerge as

so technically challenging that they are frequently ignored. The practical way of dealing with these issues begins with an understanding of DICOM.

In the following pages you will not have to wade and sift through volumes of technical specifications to get the knowledge you need. Consider it your one-stop shopping experience for a good understanding of DICOM.

David Troendle

Assistant Vice Chancellor for Information Technology

LSU Health Sciences Center,

New Orleans, LA, USA

Preface

Who Needs this Book?

If you are involved in any form of medical work, decision-making, image analysis, or research and development, you need this book. Millions of clinical practitioners greatly depend on DICOM (Digital Imaging and COmmunications in Medicine), but surprisingly few have a clear idea of what DICOM is, or even what the acronym means. As digital medicine spreads into even the most specialized areas of classical medical practice, knowing DICOM becomes increasingly imperative for ensuring the accuracy, efficiency, and reliability of any medical application or process. DICOM is an extremely powerful tool when you know how to properly implement it and can harness its power to your advantage. But it could just as easily be the bane – and even the doom – of a poorly planned clinical project. Helping you learn how to properly implement and use the features and power of DICOM was the principal reason for writing this book.

A lesser, but no less important, reason was the sheer repetitiveness of DICOM problems that I learned to deal with personally as a medical information technology administrator, researcher, and developer. To a great extent this book has evolved from my personal “DICOM diary”, which I started writing to record the problems and the solutions of my countless experiences. There also exists a long list of DICOM-related questions, inevitably asked at every single place where digital medicine is present, and there are numerous DICOM pitfalls that many still keep encountering. Although it would be too ambitious on my part to protect you from all of them, together we will try to identify and perhaps help you to avoid at least the most common DICOM traps.

Finally, the DICOM standard – being a very complex and evolving standard – is not exactly fun to read. On the other hand, many popular overviews tend to be sketchy and overly introductory. In this book, I have tried to navigate between the Scylla and Charybdis of simple to understand and important to know. This journey is not expected to be easy, but it will be rewarding. You will become familiar with the principal DICOM concepts and terminology, you will see how they fit together, and you will gain sufficient knowledge to start dealing with DICOM in all its complexity. Your next best resource after reading this book is the DICOM standard itself, and you will be prepared for it.

I hope that this educational foray will be entertaining, informative, and will change the way you look at your work. So let's go!

Oleg Pianykh

New Orleans, USA – Moscow, Russia – Boston, USA

2005–2008

Contents

PART I: INTRODUCTION TO DICOM

1	What Is DICOM?	3
2	How Does DICOM Work?	7
3	Where Do You Get DICOM from?	11
3.1	DICOM vs. Digital	11
3.2	DICOM, DICOM-Compatible, DICOM-Ready?	12
3.3	In the Middle of Nowhere	14
4	A Brief History of DICOM	17
4.1	How Did This All Get Started?	17

PART II: DICOM AND CLINICAL DATA

5	Parlez-Vous DICOM?	25
5.1	IT Boot Camp	25
5.2	Text vs. Binary	27
5.3	DICOM Grammar: Value Representations	29
5.3.1	VR Length	34
5.3.2	Characters: Foreign and Wild	35
5.3.3	Text VRs: CS, SH, LO, ST, LT, UT	36
5.3.4	Dates and Times: DA, TM, DT, AS	36
5.3.5	Numbers in Text Format: IS, DS	37
5.3.6	Numbers in Binary Format: SS, US, SL, UL, FL, FD, OB, OW, OF, AT	37
5.3.7	PN: Storing Person's Names	38
5.3.8	AE: Naming Application Entities	39
5.3.9	UIDs: Unique Identifiers	39
5.3.10	SQ: Sequencing Data Sets	40
5.3.11	UN: Representing Unknown Values	40
5.4	DICOM Data Dictionary	42
5.4.1	Standard DICOM Data Dictionary	42

5.4.2	Private DICOM Data Dictionaries	45
5.4.3	Standard DICOM Command Dictionary	46
5.5	DICOM Objects	46
5.5.1	Encoding Data Elements	48
5.5.2	Encoding Data Groups	53
5.5.3	Example: Element and Group Lengths	55
5.5.4	Encoding DICOM Data Objects	56
5.5.5	SQ: Encoding DICOM Object Sequences	57
5.5.6	Required and Optional Data Elements	63
5.5.7	Storing Image Data	64
5.5.8	Unique Identifiers	65
5.6	DICOM Information Hierarchy	67
5.6.1	Problems with Patient ID	69
5.6.2	Problems with Study, Series, and Image UIDs	71
5.6.3	Hierarchical and Relational Data	72
5.7	Modules, IODs, and Information Entities	74
5.7.1	Macro Attributes: Making it Easier	75
5.7.2	Information Modules: Basic Data Blocks	75
5.7.3	Information Entities	79
5.7.4	DICOM Information Objects	80
5.7.5	IODs and Their Instances	81
5.7.6	Learning More	83
6	Medical Images in DICOM	85
6.1	DICOM BMPs	85
6.2	Image Compression	90
6.2.1	Lossless Compression	92
6.2.2	Lossy Compression	93
6.2.3	Streaming Compression	95
6.2.4	Choosing the Right Compression Technique	98
6.3	Working with Digital Medical Images	101
6.3.1	Image Interpolation	101
6.3.2	Image Reconstructions	103
6.3.3	Grayscale Depth	106
6.3.4	Waveforms	109

PART III: DICOM COMMUNICATIONS

7	DICOM SOPs: Basic	115
7.1	Identifying Units on the DICOM Network	116
7.2	Services and Data	120
7.2.1	DIMSE Services	121
7.2.2	Simple DIMSE Example: C-Echo	122
7.2.3	Service-Object Pairs	130

7.2.4	The Verification SOP	131
7.3	Storage	132
7.3.1	C-Store IOD	134
7.3.2	C-Store DIMSE	135
7.4	Query: Find	138
7.4.1	A Few Words on Data Matching in DICOM	140
7.4.2	C-Find IOD	142
7.4.3	C-Find DIMSE	145
7.4.4	C-Cancel	148
7.5	Modality Worklist	149
7.5.1	The MWL IOD	151
7.5.2	The MWL DIMSE	152
7.6	Basic DICOM Retrieval: C-Get	152
7.6.1	The C-Get IOD	154
7.6.2	C-Get DIMSE	157
7.7	Advanced DICOM Retrieval: C-Move	159
7.7.1	The C-Move IOD	161
7.7.2	The C-Move DIMSE	162
7.7.3	C-Move vs. C-Get	164
7.8	DICOM Ping, Push, and Pull	166
7.9	Gentleman's Toolkit	167
7.10	Matching Application Roles	167
8	DICOM SOPs: Beyond Basic	171
8.1	Storage Commitment	171
8.2	Secondary Capture	172
8.3	Structured Reports	173
8.4	Encapsulated PDFs	175
8.5	Hardcopy Printing	177
9	DICOM Associations	179
9.1	Association Establishment Basics	180
9.2	Association Establishment	182
9.3	Abstract Syntax	182
9.4	Transfer Syntax	187
9.5	Application Context	190
9.6	Presentation Context	192
9.7	User information	195
9.8	Protocol Data Unit	197
9.8.1	A-Associate-RQ	199
9.8.2	A-Associate-AC	200
9.8.3	A-Associate-RJ	203
9.8.4	A-Abort	204
9.8.5	A-Release-RQ and A-Release-RP	204
9.8.6	P-Data-TF	205

9.8.7	How Associations Terminate	206
9.9	What Do I Do When it Fails?	207
9.10	Point-to-Point Spell	209
9.11	Networking: Standard and Beyond	213

PART IV: DICOM MEDIA AND SECURITY

10	DICOM Media: Files, Folders, and DICOMDIRs	221
10.1	DICOM File Format	222
10.1.1	Preamble and DICM Prefix	222
10.1.2	Group 0002: DICOM File Meta Information	223
10.1.3	Data Object	225
10.1.4	DICOM File IDs and Names	225
10.2	Special DICOM File Formats	227
10.2.1	DICOMDIR	227
10.2.2	Secure DICOM File Format	234
10.3	DICOM File Services	234
10.3.1	DICOM File Set	235
10.3.2	File Management Roles and Services	235
10.4	Grains of Salt	238
10.4.1	DICOMDIR	238
10.4.2	Media Storage	240
10.4.3	Please, Send Us Some Film!	241
10.4.4	Export and Import	241
10.5	Storing DICOM Data in PACS	242
10.5.1	File-Based PACS Storage	243
10.5.2	Database-Based PACS Storage	244
10.5.3	Mixed PACS Storage Models	245
10.5.4	Choice of Internal File Format	246
11	DICOM Security	247
11.1	DICOM Hacking	247
11.2	Securing the Workflow	249
11.3	Securing the Data	251
11.3.1	Anonymization	251
11.3.2	Encryption	256
11.3.2.1	How it All Works	256
11.3.2.2	Encrypting the Data	258
11.3.2.3	Verifying Data Integrity	258
11.3.2.4	Validating Data Origin	259
11.4	Concluding Remarks on Security	261
12	Incompatibility of Compatible	263
12.1	DICOM Conformance	263

12.2 Testing, Testing, and Yes, More Testing 264

12.3 Who Wants DICOM? 266

12.4 DICOM from a Black Box 267

12.5 “Home-Made” DICOMs 268

12.6 Open-Source DICOM 269

PART V: ADVANCED TOPICS

13 Teleradiology 275

13.1 Can I See the Image? 275

13.2 You’ve Got Mail 278

13.3 Teleradiology 280

13.4 DICOM and the WWW 286

13.5 “DICOM Email” 289

13.6 From PACS to PDAs? 292

13.7 Starting Your Teleradiology Project 295

13.8 Conclusion 298

14 Standards and System Integration in Digital Medicine ... 301

14.1 HL7: HIS and RIS 302

14.2 Integration Problems 305

14.2.1 Data Integration 307

14.2.2 Application Integration 308

14.3 IHE: Integration Profiles 309

15 Disaster PACS Planning and Management 313

15.1 What it Takes to Kill a PACS 313

15.2 Extreme PACS 316

15.2.1 Digging in the Dirt 317

15.2.2 Confidentiality and Security 319

15.2.3 Archiving 320

15.2.4 Disaster-Proof PACS Design 320

16 DICOM Software Development 325

16.1 Developing DICOM from Scratch 325

16.2 The “I just need to open this DICOM file” Project 325

16.3 Implementing DICOM 326

16.4 Learning from Others 329

17 DICOM Implementation Plans 331

17.1 Imaging Center 331

17.2 Teleradiology Center 332

17.3 Hospital 334

17.4 Image-Processing Laboratory 335

18	DICOM FAQs	339
18.1	Frequent Problems	339
18.1.1	I Want to Go Digital. How Do I Start?	339
18.1.2	How Do I Distinguish DICOM Files from Others?	339
18.1.3	I Am Trying to Open a File in My DICOM Program, and it Does Not Open	340
18.1.4	What Is DICOMDIR?	340
18.1.5	I Send a Study to My DICOM Device, but it Never Gets There	341
18.1.6	I Send a Study to My PACS Archive: it Arrives There, but Becomes Merged with Another Study (Patient)	341
18.1.7	I Need to Send This Study to a Remote Facility	341
18.1.8	I Open a DICOM Image in My Software, and it Looks Wrong	342
18.1.9	I Do Some Image Editing on My PACS, but When I Send it to Another System, the Editing Is Missing	342
18.1.10	Can I Break My DICOM Device if I DICOM-Connect it to Another?	342
18.2	Naïve Questions that Physicians Like to Ask and Salesmen Like to Brag About	343
18.2.1	Do I Get the Original Image Resolution if I Buy Your PACS?	343
18.2.2	Is Your System Web-Based?	343
18.2.3	Can You Connect to My CT Scanner (MR, Ultrasound, Positron Emission Tomography, and so on)?	344
18.2.4	Do You Have 3D Imaging (Perfusion, Diffusion, Volume Measurements, and so on)?	344
18.2.5	Does Your System Support Multitasking (e.g., Multithreading, 64-bit Processors)?	345
18.2.6	Can I Use Your System for Teleradiology?	345
18.2.7	How Fast Can You Transfer the Images?	346
18.2.8	How Much Data Can I Store in Your PACS?	346
18.2.9	Is Your DICOM Software Secure?	347
18.2.10	Do You Have Full-Fidelity DICOM?	347
	Appendix	351
A.1	DICOM Transfer Syntaxes	351
A.2	DICOM SOPs	353
A.3	C-Find Bytes	359
	References	379
	Subject Index	381

Abbreviations

3D	Three-Dimensional
AAPM	American Association of Physicists in Medicine
ACR	American College of Radiology
AE	Application Entity
AET	Application Entity Title
AS	Age String
ASP	Application Service Provider
AT	Attribute Tag
blob	Binary Large Object
BMP	Bitmap
CAD	Computer-Aided Diagnosis
CR	Computed Radiography
CS	Code String
CT	Computed Tomography
DA	Date
DICOM	Digital Communications in Medicine
DICOM UL	DICOM Upper-Layer Protocol
DIMSE	DICOM Message Service Elements
DIMSE-C	Composite DIMSE
DIMSE-N	Normalized DIMSE
DS	Decimal String
DT	Date/Time
ECG	Electrocardiogram
EPR	Electronic Patient Record
FD	Floating Point Double
FDA	US Food and Drug Administration
FL	Floating Point Single
FSC	File Set Creator
FSR	File Set Reader
FSU	File Set Updater
Gb	Gigabit
GB	Gigabyte
Gbs	Gigabits per Second
HIPAA	Health Insurance Portability and Accountability Act
HIS	Hospital Information System
HTML	Hypertext Markup Language
HTTP	Hypertext Transfer Protocol
ID	Identification Number
IE	Information Entity
IHE	Integrating the Healthcare Enterprise
IOD	Information Object Definition
IP	Internet Protocol

IS	Integer String
ISO	International Organization for Standardization
IT	Information Technology
JND	Just Noticeable Difference
JPEG	Joint Photographic Experts Group
Kb	Kilobit
KB	Kilobyte
Kbs	Kilobits per Second
LO	Long String
Mb	Megabit
MB	Megabyte
Mbs	Megabits per Second
MCH	Message Control Header
MIME	Multipurpose Internet Mail Extensions
MOD	Magneto-Optical Disk
MR	Magnetic Resonance
MWL	Modality Worklist
NEMA	National Electrical Manufacturers Association
NM	Nuclear Medicine
OB	Other Byte String
OF	Other Float String
OW	Other Word String
PACS	Picture Archiving and Communication System
PDF	Portable Document Format
PDU	Protocol Data Units
PDV	Protocol Data Value
PN	Person's Name
RIS	Radiology Information System
RLE	Run-Length Encoding
ROI	Region of Interest
RSNA	Radiological Society of North America
SC	Secondary Capture
SCP	Service Class Provider
SCU	Service Class User
SH	Short String
SL	Signed Long
SMTP	Simple Mail Transfer Protocol
SOP	Service-Object Pair
SR	Structured Report
UL	Upper Layer
URL	Uniform Resource Locator
VPN	Virtual Private Network
WAN	Wide-Area Network
WG	Working Group
WWW	World-Wide Web
XML	Extensible Markup Language

PART I: INTRODUCTION TO DICOM

Chapter 1

What Is DICOM?

“When working toward the solution of a problem, it always helps if you know the answer.”

Murphy’s Law

You can walk with this question into the most modern, digital, state-of-the-art hospital and spend hours looking for someone who could answer it correctly. We all get used to buzz words and acronyms, and rarely think about their meanings. Unfortunately, nothing distances you more from success than not knowing what you are dealing with!

DICOM stands for Digital Imaging and COmmunications in Medicine and represents years of effort to create the most universal and fundamental standard in digital medical imaging. As such, it provides all the necessary tools for the diagnostically accurate representation and processing of medical imaging data. Moreover, contrary to popular belief, DICOM is not just an image or file format. It is an all-encompassing data transfer, storage, and display protocol built and designed to cover all functional aspects of digital medical imaging (which is why many view DICOM as a set of standards, rather than a single standard). Without a doubt, DICOM truly governs practical digital medicine.

Another important acronym that seemingly all DICOM companies plug into their names is PACS (Picture Archiving and Communication Systems). PACS are medical systems (consisting of necessary hardware and software) designed and used to run digital medical imaging. They comprise digital image acquisition devices (modalities – such as computed tomography (CT) scanners, or ultrasound), digital image archives (where the acquired images are stored), and workstations (where radiologists view the images). When you play with your digital camera (modality), store the images on your computer (archive), and send them to your friends (reviewers), you use the exact same model. Of course, PACS take the model to a much more complex level (Fig. 1).

PACS are directly related to DICOM. Their functionality is DICOM-driven, which ensures their interoperability. For that reason, any PACS device or software comes with its own DICOM Conformance Statement, which is a very important document explaining the extent to which the device supports the DICOM standard. In essence, PACS bring the DICOM standard to life.

One can hardly imagine modern-day digital medicine without DICOM and PACS. The DICOM standard – conceived over 20 years ago – plays an integral

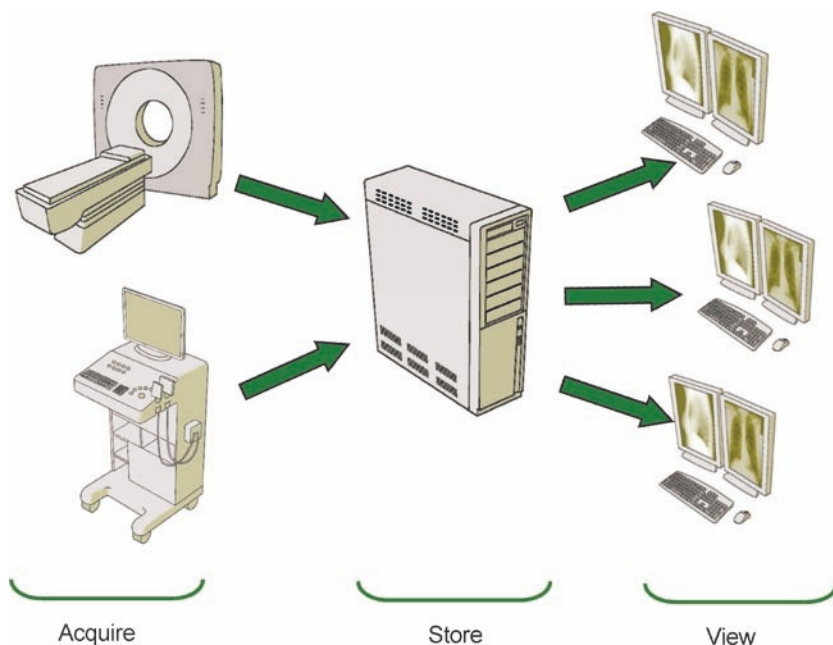


Fig. 1 Major Picture Archiving and Communication System (PACS) components. Image acquisition devices (modalities) store images on a digital archive. From there images are accessed by radiologists at the viewing workstations

role in the digital medicine evolution, ensuring the highest diagnostic standards and the best performance. DICOM has truly shaped the landscape of contemporary medicine by providing:

1. *A universal standard of digital medicine.* All current, digital image-acquisition devices produce DICOM images and communicate through DICOM networks. Current medical workflow is implicitly controlled by a multitude of DICOM rules, which will be reviewed in this book.
2. *Excellent image quality.* For example, DICOM supports up to 65,536 (16 bits) shades of gray for monochrome image display, thus capturing the slightest nuances in medical imaging. In comparison, converting DICOM images into JPEGs or bitmaps (BMP), always limited to 256 shades of gray, often makes them impractical for diagnostic reading. DICOM takes advantage of the most current and advanced digital image representation techniques to provide the utmost diagnostic image quality.
3. *Full support for numerous image-acquisition parameters and different data types.* Not only does DICOM store the images, but it also records a multitude of other image-related parameters such as patient 3D position, physical sizes of objects in the image, slice thickness, image exposure parameters, and so on. These data immensely enrich the informational content

of DICOM images, and facilitate the processing and interpretation of the image data in various ways (for example, creating 3D images from several sequences of two-dimensional CT slices).

4. *Complete encoding of medical data.* DICOM files and messages use more than 2000 standardized attributes (DICOM data dictionary) to convey various medical data from patient name to image color depth, to current patient diagnosis. These data are often essential for accurate diagnostics, and capture all aspects of the current radiology.
5. *Clarity in describing digital imaging devices and their functionality – the backbone of any medical imaging project.* DICOM defines medical device functionality in very precise and device-independent terms. Working with medical devices through their DICOM interfaces becomes a very straightforward process, leaving little room for errors.

At the time this book was written, the DICOM standard consisted of 16 volumes (from 1–18, volumes 9 and 13 being retired) known as parts, and traditionally numbered from PS3.1 to PS3.18.¹ The last publicly available revision of the standard, performed in 2007, was used.²

¹ Number 3 representing DICOM 3.0, the current version of the standard.

² See DICOM home page at NEMA's Web site, <http://medical.nema.org>. When this book was in print, DICOM version 2008 was released.

Chapter 2

How Does DICOM Work?

*“Everything in life is important,
important things are simple, simple things are never easy.”*

Murphy’s Law

To introduce order into the complex medical environment, DICOM uses its own lingo, based on its model of the real world (DICOM information model). Here is that model in a nutshell.

All real-world data – patients, studies, medical devices, and so on – are viewed by DICOM as objects with respective properties or attributes.³ The definitions of these objects and attributes are standardized according to DICOM Information Object Definitions (IODs). Think about IODs as collections of attributes, describing each particular data object. A patient IOD, for example, can be described by name, medical record number (ID), sex, age, weight, smoking status, and so on – as many attributes as needed to capture all clinically relevant patient information. In a broader sense, a patient (just like any other DICOM object) is the set of attributes of which he consists, as you can see on Fig. 2. DICOM maintains a list of all standard attributes (more than 2000 of them), known as the DICOM data dictionary, to ensure consistency in attribute naming and processing. For example, our patient attributes – name, date of birth, sex, and so on – are also included in the DICOM Data Dictionary. All DICOM attributes are formatted according to 27 value representation (VR) types, corresponding to dates, times, names, and so on.

As soon as the data is captured as DICOM data attributes, it can be transmitted and processed between various DICOM devices and software (Application Entities (AEs), as they are known in DICOM). DICOM represents this processing with a service-rendering model: DICOM applications provide services to each other. Because each service usually involves some data exchange (typically performed over a computer network), it becomes natural to associate particular service types with the data (IODs) that they process. DICOM calls these associations Service-Object Pairs (SOPs), and groups them into SOP Classes. For example, storing a CT image from a digital CT scanner to a digital

3 If you have an IT background, you will certainly recognize object-oriented design.

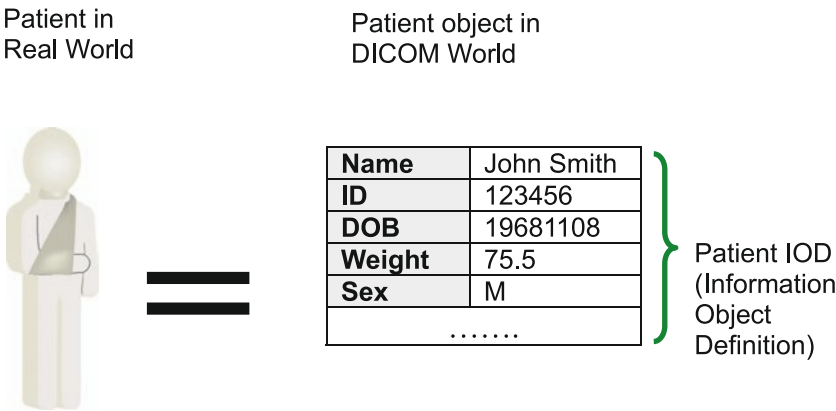


Fig. 2 From real data to DICOM Information Object Definitions (IODs). Each IOD is a collection of attributes

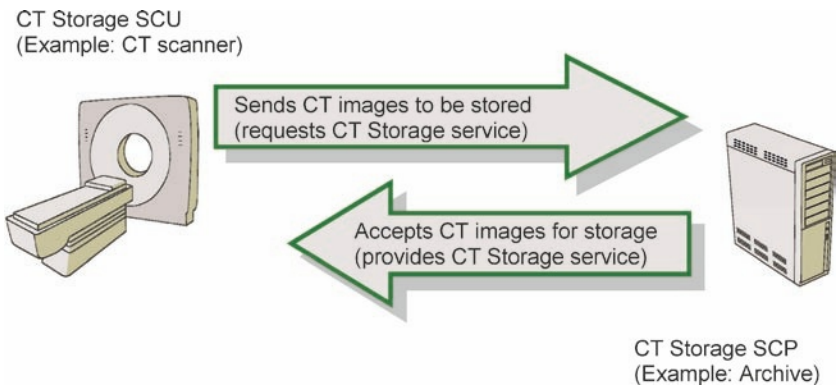


Fig. 3 DICOM services

PACS archive corresponds to the CT Storage SOP, as shown on Fig. 3. In this particular example, the CT image represents the DICOM IOD (DICOM data object).

The CT scanner requests the storage service from the archive, and the archive provides the storage service to the scanner. To differentiate between service requestors and service providers, DICOM calls the former Service Class Users (SCUs) and the latter Service Class Providers (SCPs). In the same CT example, the CT scanner acts as the CT Storage SCU, and the digital archive as the CT Storage SCP.

Each DICOM network data exchange between SCU and SCP peers is called association. Consequently, each network transfer begins with an Association

Establishment–DICOM handshake, when the two connecting applications exchange information about each other. This information is called the Presentation Context; if the two applications can match their contexts, they can connect and start SCU-SCP processing.

Because hundreds of DICOM devices and applications are produced by hundreds of DICOM manufacturers, each DICOM unit will be accompanied by its own DICOM Conformance Statement from the manufacturer. This statement explains which SOPs (services) the unit supports, and to what extent (SCU, SCP, or both). The DICOM Conformance Statement is your most essential roadmap for any DICOM-related project. Obtain it from the manufacturer ahead of time and read it carefully. For example, if you buy a digital archive that supports only CT Storage SCU (does not support CT Storage SCP) you won't be able to store CT images in it. The archive won't be able to provide the CT storage service.

This brief summary reflects the core DICOM functionality, and as you can see, it is quite straightforward. In fact, understanding the theory of DICOM is easy; dealing with DICOM in real life is often the challenge. Most of this book is committed to helping you to meet that challenge.

Chapter 3

Where Do You Get DICOM from?

“If anything can go wrong, it will.”

Murphy’s Law

The DICOM standard itself is free and can be found on the official DICOM home page (<http://medical.nema.org>), which is maintained by the National Electrical Manufacturers Association (NEMA). However, from a practical perspective, you usually deal with DICOM implementations in devices and software. Currently, hundreds of these products are on the market, constantly competing, transforming, and vying for your attention. This natural selection, however, only adds another layer of commonly misunderstood and confusing concepts to the original complexity of the DICOM standard.

Moreover, the question “Are you completely DICOM-compatible?” can never be answered with a simple “Yes” or “No”. In fact, this question is just plain inaccurate. All DICOM devices and software implement only specific parts of the DICOM standard required for their functionality. As a result of this variety, implementing DICOM workflow has become one of the most critical and treacherous phases in organizing any medical imaging practice. How do you introduce DICOM into your practice and what should you avoid? Well, let’s start with the basics.

3.1

DICOM vs. Digital

Like all human beings, we perceive the world around us in analog mode – that is, as continuous shapes and shades. Computers on the other hand function in digital mode – they store and process images dot-by-dot, number-by-number. DICOM, as its very name suggests, also deals only with digital images. Therefore, acquiring or converting images into digital format is the very first step required for any DICOM implementation.

All contemporary, image-acquisition modalities provide a digital image output. Such modalities include CT, magnetic resonance imaging (MR), ultrasound (US), nuclear medicine (NM), and more. In fact, some of those devices have always been digital simply because of the nature of their image-acquisition

method, while others were complemented with digitizing circuits as technology progressed.

The problem, however, is that while DICOM implies digital, digital does not necessarily guarantee DICOM. You could very well purchase a digital medical imaging device that has no DICOM support whatsoever.

How would you know if you can use DICOM with your medical imaging device? It's easy. As already mentioned, any DICOM device should come with a DICOM Conformance Statement.⁴ The conformance statement, in many cases, is more important than the device's user manual and should be supplied by the device manufacturer to outline the DICOM functions supported by the device. For this very reason, the statement should leave no room for errors or false assumptions about the device's functionality.

You should be conjuring the guardian spirits of DICOM Conformance Statements any time you plan or perform any DICOM-related installation. Always ask for it, and always make sure that you have the correct version for your model. Even the same device or software can come in a variety of models and revisions, and your conformance statement should correspond to the model/version that you are using.

3.2

DICOM, DICOM-Compatible, DICOM-Ready?

Let's say that you have identified your imaging needs and you are ready to purchase a DICOM unit. Is there anything you need to know about before writing that check? Definitely. Buying a DICOM-compatible device or software application is just like having dinner in an expensive restaurant. Your device manufacturer presents you with a long menu of entrées, most of which have fancy names and exorbitant prices. And each entrée can be complemented by various, optional side dishes. Choose carefully! A device's ability to run the DICOM standard could very well be one of these optional sides. Many clinical practitioners have committed this very mistake: skipping the optional and not ordering what they really should have and found themselves eating crow in the end.

You might ask: "Why should we pay for DICOM functionality as an additional DICOM device option, when having a DICOM device without DICOM is meaningless?" Well, it might be meaningless to you, but it surely is not meaningless to the device manufacturers. To better comprehend this apparent paradox, let's recall what we learned in the previous section.

⁴ Our references to this document in this book might seem infinite; however, this is due to its utmost importance in the DICOM application roles.

All current, medical image-acquisition devices are digital, which does not necessarily mean that they inherently provide DICOM functionality. Device manufacturers can implement their own proprietary protocols for digital image acquisition, storage, and display. Those manufacturer-dependent protocols will suffice to operate the device as a stand-alone unit, or connected to the other units from the same manufacturer. Thus, for the manufacturers, the use of their proprietary protocols involves the issues of convenience and marketing. Convenience because the manufacturers can do anything they want to implement the device functionality internally – they are not limited by any standard or any particular requirement; marketing because proprietary standards make you dependent on your manufacturer – you simply cannot connect to another manufacturer’s equipment or software.

This might sound like just another conspiracy theory, but pragmatically, any proprietary format inevitably segments the medical device market. This will be discussed in more detail at the end of this book (see Chap. 12).

The bottom line is that in many cases, you must explicitly purchase DICOM functionality as an option (in addition to the DICOM device purchase itself). If you do not purchase the DICOM functionality option, you end up with what some manufacturers label as a “DICOM-Ready” device, a device that could have been running DICOM, if you had paid for its DICOM functionality option.

Another way of looking at this problem is to realize that, in its very essence, DICOM is a device-interfacing standard. It connects the devices externally, but it is not meant to drive them internally. DICOM provides a standard for device output (which also ensures that various devices can be connected to each other). When you purchase a DICOM option, you pay for this conversion, you pay for uniformity, and you pay for the ability to export device data in DICOM format.

If you are serious about your medical imaging workflow, all DICOM options on your manufacturer’s price list should be treated as mandatory.

What should you do if any of these options are missing? Heed these words of wisdom: despite the urgency and possible sense of guilt, never attempt to fix this problem yourself. If you have a DICOM-Ready device that is lacking DICOM options, the only way to remedy the problem is to contact the device manufacturer. Resolving all DICOM issues with the manufacturer should become your standard approach under any circumstance. Not only does it deliver the best functional solution (even if you have to pay for it), but it also preserves the device warranty and ensures that no harm will be done to the unit.

A similar DICOM-compatibility problem can arise with an old imaging unit. As we will see in Chap. 4, the DICOM standard has been around for a long time and has evolved considerably. As DICOM devices age, they need to be updated with current DICOM software from their manufacturers. In general, this is the same approach but with one new catch: the original device manufacturers might not exist anymore. If this is the situation, you will more likely have one of the following remedies:

1. *Your original DICOM unit manufacturer was purchased by another, which will help you with the unit support.*
2. *Your original DICOM unit manufacturer went out of business.* In this case, often times the unit support is transferred to another DICOM provider. Many discontinued DICOM units from out-of-business companies are still sold refurbished and supported by other manufacturers.

The rule of thumb is that digital medical devices that are more than 10 years old should be replaced. In addition to advances in the DICOM standard, the entire digital medicine technology has made great strides. After a certain age the older units not only start looking primitive, but they also lack many features and functions, some of which are even considered standard with current technology. When this becomes the case, do not keep patching these dinosaurs; you will do much better replacing them with contemporary models.

3.3

In the Middle of Nowhere

There remains one more common case of providing DICOM functionality: medical devices that were never meant to support the DICOM standard. Despite the rapid progress in digital medicine technology, and the growing DICOMization of the medical imaging workflow, DICOM-free units are still numerous and can come from several sources:

1. Digital medical devices manufactured with no DICOM interface.
2. Generic nonmedical devices (digital or analog).
3. Analog medical devices with no digital output.
4. Old, pre-DICOM medical devices.

The last case of pre-DICOM units is the least interesting and will be discussed in the following chapter on DICOM history. In general, if your device is older than DICOM (which is quite old in itself), it is definitely the time to consider buying a new one.

The first case on our list is much more typical: contemporary digital medical imaging devices (such as many popular film digitizers) that were never meant to have a DICOM interface. This usually happens when device manufacturers want to keep their distance from the medical imaging domain; either because the manufacturer works in other areas/markets, or because it considers medical imaging too complex and troublesome for a stronger commitment. For example, if a CT scanner must be DICOM-compliant, a simple flatbed scanner used elsewhere might not.

The problem with digital non-DICOM devices is bigger than their lack of DICOM output. They simply do not fit well into the clinical workflow. For example, each DICOM image must contain the patient name and ID; this is ex-

tremely important for accurate image routing and identification. If you have a film digitizer that scans films into plain, non-DICOM digital image formats (such as BMP), tagging them with the patient or study information is simply impossible. Non-DICOM image formats cannot support DICOM tags. Diagnostic image quality (excellent in DICOM and greatly varying in conventional image formats) is another issue. While DICOM modalities store all clinical data automatically in DICOM image tags, their less-advanced and more generic non-DICOM counterparts require human assistance, which often leads to accidental errors and data loss.

With all of this in mind, if you still want to buy a digital medical device with no DICOM interface, at least take the following precautions:

1. Try to buy the device through a DICOM manufacturer, which might resell the device with its own DICOM interface.
2. Do not make a non-DICOM device the centerpiece of your clinical workflow.

The same logic applies to using generic, nonmedical devices (digital or not). For example, many dermatologists would like to use off-the-shelf digital cameras to record images of their patients' skin patterns, infections, and other conditions (which makes perfect sense clinically). Clearly, off-the-shelf digital cameras have no DICOM functionality whatsoever, and likely never will. Therefore, if you want to incorporate those images into your PACS or medical imaging workflow in general, you face the same problem of DICOM conversion and image identification. Because those devices were never meant to be medical, your chances of getting any DICOM assistance from their manufacturer or any DICOM manufacturer are beyond slim. Instead, look for image-importing options in your existing DICOM software. Many current DICOM workstations and file viewers offer various options for DICOM import, enabling you to convert a plain, conventional digital image (BMP, JPEG, and such) into a standard DICOM image. Once again, you will have to manually enter the missing information (patient name, ID, study date, and so on), but you will be rewarded with the ability to include this image into your clinical cycle. In a more general sense, the same approach works for non-DICOM image files, wherever they might come from.

Many medical devices are still analog and could stay analog for some time (for example, nonimaging units such as electrocardiogram (ECG), or even certain modalities such as ultrasound are sometimes not equipped with digital interfaces). As we know, a device must be digital to be DICOM. Therefore, analog images must be digitized. Still images can be converted into digital format and video can be broken into digital image frames. The task of digitization and DICOM conversion is commonly performed with DICOM converters, which are small box-like devices capable of converting analog images, video, and cine loops into digital, DICOM-compliant images. Moreover, DICOM converters can often interface with standard hospital information systems (to obtain pa-

tient data instead of relying on error-prone manual entry) and can send converted images to PACS and other DICOM devices, thus supporting DICOM networking in addition to plain image conversion. DICOM converters are primarily marketed for legacy analog imaging systems, but will do the job with any analog imaging device in general. Another good thing about them is that some DICOM converters are actually smart enough to work with certain proprietary imaging formats from the main medical producers and can convert those formats to work with DICOM as well. In addition to DICOM converters, more and more current DICOM/PACS products have begun to offer advanced image export options, accommodating at least digital (but non-DICOM) images.

To conclude, it is still possible to create a DICOM workflow from generic, somewhat crude, and non-DICOM-compliant imaging devices. Nevertheless, as mentioned earlier, image quality and bookkeeping can be problematic. In addition, the need for human interaction makes such solutions slower and more prone to error. Therefore making DICOM out of nothing is more appropriate for low-volume, marginal imaging solutions, or as a temporary solution in the midst of your transition from a legacy system to a contemporary PACS. However, steady, industrial medical imaging workflow definitely needs to be based on 100% DICOM-compliant devices.

Chapter 4

A Brief History of DICOM

“Every solution breeds new problems.”

Murphy’s Law

Because the DICOM standard is some 20 years old, its history has become an integral part of its being; and knowing the history of DICOM can help answer many current questions. Moreover, despite having undergone frequent revisions, the standard has never truly been revolutionized. It has continually evolved, adjusting itself to current practices, yet preserving many of its original historical features.

The natural process of DICOM device manufacturing, selling, and using spans a cycle of several years (modalities are expensive, and hospital administrators are typically conservative and budget-conscious, trying to get the most out of equipment), sometimes even to the point of keeping things until they fall apart. Should we also mention all discontinued, refurbished, and simply old units that many practices still purchase as the most affordable alternative? This creates an environment in which drastic updates are not really welcome, and compatibility with older equipment (and, consequently, older DICOM) becomes a must.

In brief, if you work in the current, multifaceted clinical environment you will have to work with multigeneration equipment (DICOM-compliant and not), and you are bound to make occasional “archeological” discoveries, finding yourself dealing with many layers of DICOM history.

4.1

How Did This All Get Started?

The standard was conceived in 1983 by a joint committee formed by the American College of Radiology (ACR), and the National Electrical Manufacturers Association (NEMA).⁵ The primary goal was to develop a standard that would

⁵ Part PS3.1 of DICOM standard includes a brief historical overview of DICOM.